

DAY – 12

Improve 3D Try-On Research

In this task, I continued exploring the concept of 3D Virtual Try-On systems to understand how clothes can be digitally fitted on a virtual human model in a more realistic way compared to 2D try-on systems. Unlike 2D virtual try-on, where clothing is simply overlaid on an image, 3D try-on technology attempts to simulate real garment fitting on a three-dimensional body model. This helps improve realism, body fitting accuracy, and cloth movement behavior.

1. 3D Avatar Generation

One of the major components of a 3D try-on system is avatar generation. A virtual avatar is created to represent the user's body digitally. This avatar is usually generated using body measurements, pose estimation, or image-based body analysis.

The purpose of creating a 3D avatar is to provide a structure where garments can be tested and visualized before actual purchase.

- Key areas explored in avatar generation:
- Creating a digital body representation
- Estimating body proportions from images
- Improving avatar size and fitting accuracy
- Matching clothing dimensions to user shape



2. Body Mesh Fitting

After avatar creation, the next step is body mesh fitting. In a 3D try-on system, clothing is not simply placed on top of an image. Instead, garments are fitted onto a virtual body mesh. A body mesh is a structured 3D model made up of connected points and surfaces representing the shape of the human body.

The fitting process adjusts the garment according to body proportions such as shoulder width, chest size, waist alignment, and body posture. This helps improve realism and avoids incorrect clothing placement.

Important aspects of body mesh fitting include:

- Shoulder and torso fitting
- Body size adjustment
- Garment scaling based on measurements
- Improved fitting realism for different body types



3. Cloth Simulation Realism

Cloth simulation is another important concept in 3D virtual try-on systems. In real life, clothes move naturally according to body posture, gravity, and fabric type. A realistic try-on system attempts to recreate this behavior digitally.

Cloth simulation helps garments behave naturally instead of appearing rigid or fixed. For example, jackets fold near the elbows, shirts stretch depending on movement, and loose clothes react differently compared to tight garments.

Areas explored under cloth simulation include:

- Cloth bending and stretching
- Gravity effects on garments
- Fabric movement behavior
- Wrinkle and fold generation

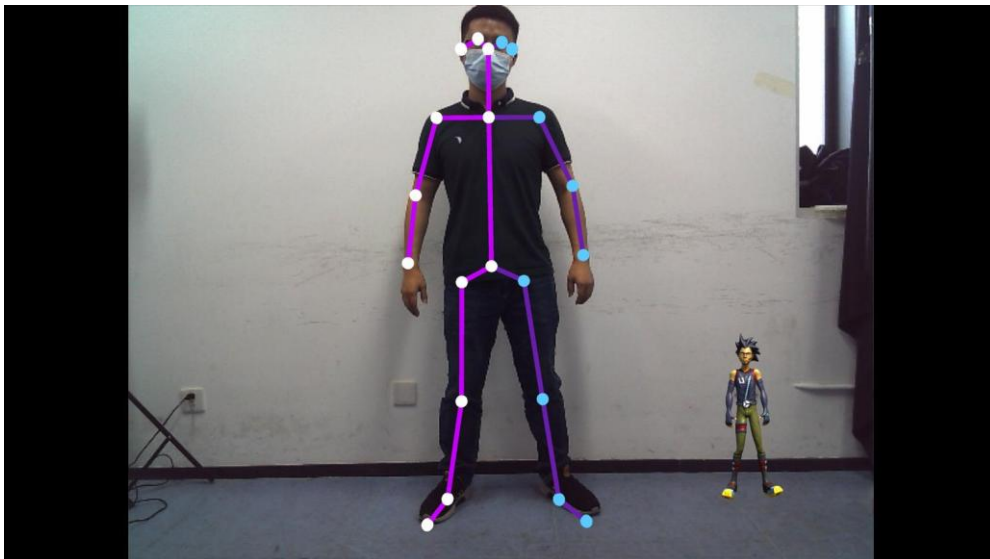
4. Avatar Movement Tracking

Another important part of 3D try-on research is avatar movement tracking. Since human movement constantly changes body posture, a try-on system should update garment fitting dynamically.

Body landmarks and skeleton tracking are used to understand motion and reposition clothes according to movement. This helps simulate realistic try-on experiences during walking, turning, or arm movement.

Areas explored in movement tracking:

- Pose estimation and body landmark tracking
- Skeleton movement detection
- Dynamic cloth adjustment during movement
- Realistic garment response to body posture



Through this task, I explored the working principles of 3D Virtual Try-On systems and understood how avatar generation, body mesh fitting, cloth simulation, and movement tracking contribute to a more realistic virtual fitting experience. Compared to 2D methods, 3D try-on systems provide better realism and personalization but require higher computational resources and more advanced modeling techniques.

These can be done using blender and mediapipe.